INTERVIEW-PREP PACKET 2 OF 5 · PER-INTERVIEWER · BUSINESS-PROCESS LENS

Amazon NACF Supply Chain loop — Packet 2: Ryan Hoerger

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

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Wednesday, October 29, 2026 · 12:00 pm PDT · 55-minute video interview

SHADOW INTERVIEWER: SANDY HSIAO (A SECOND NOTE-TAKER — SAME PACE, SAME NUMBERS; EXPECT ONE QUESTION FROM THE SHADOW NEAR THE END)

> **Strategy in one line.** The most quantitative interviewer on the panel. Tell two Plan-process redesigns with the math visible and every input labeled: S6 (honest supply definition → shortage alert a lead-time early → $2M released under a two-signal rule) and S8 (brochure lead times → measured lane distributions → p99 promise → ~$1.5M released with zero stockouts). The two programs' dollars never meet. In his language (public-profile inference below): S6 is positioning safety stock where measured errors justify it; S8 is positioning the promise at p99 per lane.

LP focus — my guess, not confirmed: Dive Deep · Are Right, A Lot · Learn & Be Curious — with a mentorship angle (his public About line names "a culture of mentorship"; expect one people question). Amazon does not disclose the assignment; the guesses are reasoned from the interviewer's title, the JD and his public profile in "Read me first" (public-profile inference, labeled as such) and should be held loosely — Insist on the Highest Standards remains a live alternate via the 99% bar in S8.

Shadow etiquette: address both; the shadow is usually a new interviewer being calibrated — same numbers, same pace; if the shadow asks the last question, answer it with the same discipline (phase asked, then stop).

Canon: amazon/amazon-l6-scm-master-packet.html (§1.2 roster, §4.1 ledger). Every number below is that packet's value; every dollar carries its basis label (record · user anchor · modeled · priced).

Read me first

The interviewer's lens (inference from title, JD and public profile — the profile parts labeled)

Inference from the title. "Placement Analytics" is the analytical arm of inventory placement — deciding where inventory should sit in the network so that the promise holds at the lowest landed cost. A manager in that function lives on forecast error, lead-time variability, safety-stock policy, service levels and the data-quality underneath them. He will be the most quantitative interviewer on the panel and the most likely to ask "how was that measured?", "what was the denominator?", and "what is the softest input in your model?" He is also the one most likely to recognise a textbook formula being passed off as a program artifact — so the honesty labels matter more here than anywhere.

What that means for the answers. Two Elementum planning stories, told as Plan-process redesigns with the math visible: the shortage-alert / real-time supply-demand view (S6 — forecast accuracy 70 → 85%, $2M safety stock released under a two-signal rule) and the measured-lane lead-time program (S8 — promise at p99 on 200 lanes, 580K units / ~$1.5M released with zero stockouts). Both end on *capital released on evidence with service held* — asset utilization and reliability in the JD's words. Keep the two programs' dollars apart in every sentence. A shadow (Sandy Hsiao) is on the call: same pace, same numbers, expect one question from the shadow near the end.

What his public profile adds (public-profile inference, labeled)

The profile (public-profile inference — directional, not verified biography). Manager, Transfer/Placement Analytics at Amazon (Jul 2025–), New York; before that Senior Business Intelligence Engineer at Amazon (Dec 2024–Jul 2025); analytics manager at Nike (2021–24); analyst at Cornerstone Research (2016–20) — an economics-consulting shop where the work product is expert-testimony-grade statistical and financial analysis; Chicago Booth MBA (Siebel Scholar); Duke. His own About line: "rigorous statistical and financial analysis… a blend of data analysis and writing skills… a culture of mentorship." **Quoted for framing only — never referenced in the room; none of it enters my stories.**

What that means for the answers. Expect *definition-level probing*: MAPE versus bias and which one steered; the grain (SKU-family? per lane?); the window and refresh; what "99% lead-time accuracy" tolerates in days. Three such definition Q&A are drilled into the S6/S8 probing below. Second, *answer style: crisp and memo-like* — claim, evidence, stop; he writes for a living, and a long wind-up reads as evasion. Third, speak *placement*: S6 is positioning safety stock where the measured errors justify it; S8 is positioning the promise at p99 per lane — his daily work is deciding where inventory sits versus where it is promised. Fourth, the mentorship line in his About suggests one people probe — answer it with planners grown into owners (S6's measurement handover; S8's drift flag owned by planning), not with org charts.

Three guessed LP focus areas, with rationale

> **Labeled guess.** Amazon assigns 2–3 Leadership Principles per interviewer and does not disclose them. These three are inferred from the title, the JD and the public profile above (public-profile inference); if the first question lands on a different LP — Insist on the Highest Standards via the 99% bar, or Frugality via the capital releases — the twist maps under each story carry the pivot.

GUESSED LP	WHY THIS INTERVIEWER WOULD CARRY IT	STORY TO OPEN · BACKUP
Dive Deep	An analytics manager tests whether the candidate goes to the data grain — measured lane distributions instead of brochure values; the definition of "supply" before any screen. S8 is the Dive Deep primary in the master map.	S8 (open) · S6 (the supply definition, the MAPE/bias measurement)

GUESSED LP	WHY THIS INTERVIEWER WOULD CARRY IT	STORY TO OPEN · BACKUP
Are Right, A Lot	Placement decisions are judgment under uncertainty; he will probe how the candidate was right against the room's belief and how the decision was made safe (release rule, watch window, fat-tail exception).	S8 (measured vs brochure; p99 chosen by planning on the asymmetry) · backup: S6 (refusing unfirmed MRP orders as supply)
Learn & Be Curious (with a mentorship angle)	A Cornerstone-trained analyst with a Booth MBA who writes about rigor and mentorship (public-profile inference) will test whether the candidate learns at the definition level and grows people at the mechanism level: the MIT toolkit acquired to design what I had been implementing — labeled toolkit, never day job — and planners grown into owners of the measurement, the drift flag and the refresh cadence. Insist on the Highest Standards stays live as the 99% bar before any release and the two-signal rule before any buffer came out.	S8 (99% bar; drift flag owned by planning) · S6 (measurement handed to planners)

Answer strategy — business first, systems as enablers

- **Open with the process problem and its cost, not the system.** First sentence names the broken process (Plan, Source, Make, Deliver or Return), who it hurt (customer, Finance, the floor) and the baseline number. The platform appears only as the enabler of the redesign, in one clause.
- **Use the JD's four outcome words as the results frame:** reliability (did the promise hold?), responsiveness (how much earlier did we see it?), cost efficiency ($ with its basis label), asset utilization (capital, labor-hours, capacity released).
- **Action = operating-model redesign in gated phases** — Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each spoken as entry bar · what I delivered · exit bar. Say the gate that fired, not the meeting that happened.
- **Every dollar carries its basis word in the same breath** — record · user anchor · modeled · priced. A record dollar is said flat; a modeled dollar is said with its inputs. Never promote a modeled number to record.
- **Program-lead voice.** I owned definitions, rules, gates and the signature chain; engineers, SIs and vendor teams configured and built to the spec I signed. The only platform personally configured in canon is Manhattan DOM.
- **The pitch ends on the primary metric.** Then stop. Silence after the pitch is the interviewer filling their STAR worksheet; their probes are the second half of the answer.
- **For Placement Analytics specifically:** lead with the planning parameter that was wrong (brochure lead time; unfirmed orders counted as supply), the decision it corrupted (buffer size, shortage response), and the rule that released capital safely. Show one formula only when asked, labeled as the MIT toolkit illustration — the data team built the distributions; planners sized the buffers; Finance signed.
- **Answer style for this room: crisp, memo-like.** Claim, evidence, stop. His background (Cornerstone rigor; a writing-heavy About line — public-profile inference) means a long wind-up reads as evasion; the one-breath basis label ("record", or "modeled, on illustrative inputs") does more credibility work here than any extra sentence.
- **Speak placement.** Frame S6 as positioning safety stock — buffer placed only where measured forecast and lead-time errors justify it — and S8 as positioning the promise at p99 per lane, recomputed on a drift flag. His team decides where inventory sits versus where it is promised (public-profile inference); that is the exact shape of both stories.
- **Keep S6 and S8 apart:** $2M (S6, forecast-driven, two-signal rule) and ~$1.5M / 580K units (S8, lead-time-driven, 99% bar) are different programs — never add them, never swap their mechanisms. "Six sites" —

never "six NA sites"; never paste S6's "3 months" onto the six-site ingest (that is S7, not in this packet).

Loop-consistency rule. One question, one story; never the same story to two interviewers — interviewers compare notes. Held for other packets (do not spend here): S11/S12 (Tripathy) · S2/S18 (Goforth) · S1/S3 (Lan) · S16/S15/S10/S14 (Josyula). S7 (CM BOM) and S9 (EDI Pareto) are unallocated master-packet backups — pointers only, and no BOM-explosion content.. If pushed for "another example" on an LP, use the named backup in the question bank or the backup bench at the end of this packet, never a story allocated to another interviewer.

How to read the "L6 level evidence" block under each story. Amazon's L6 functional bar (user-supplied rubric): *Ambiguity* — strategy or network undefined; I select the stakeholders, set the goals, deliver independently. *Scope & influence* — team-level scope; influences one to two levels up. *Execution* — tactical and strategic; simplifies or extends; applies best practices; force-multiplies; mitigates long-term risk; makes the short-versus-long-term trade-off explicit. *Impact* — long-term team goals. *Process improvement* — operational-excellence best practices institutionalized. *Complex* — visible risks, roadblocks and constraints mitigated through expertise, judgment and influence to consensus. Overview themes the block is tuned to: designing and modeling the supply chain rather than working in it; developing new processes for others to own; root-cause mechanisms that resolve trending issues at scale; recommendations to leadership that outline competing metrics. The block sits right after Results so the evidence is on the page when the probing starts.

The stories — business-first, canon numbers

Two Elementum planning stories from the same six-site manufacturer. Each: 60-second pitch → Situation/Task → Action in gated phases → Results → Probing tuned to a placement-analytics manager → the key math → Twist map.

S6 — Elementum — one real-time supply/demand view across eight data domains: a shortage visible a lead-time early, forecast accuracy 70% → 85%, $2M of safety stock released under a two-signal rule

LPs this story can serve here: Dive Deep (the definition of supply; MAPE/bias at the decision grain) · Earn Trust (the two-signal release rule signed with Planning and Finance) · Deliver Results (third — alert live in 3 months) · Think Big (one view over eight domains) · Are Right, A Lot (backup — refusing unfirmed orders as supply)
Process family: Plan — S&OP / shortage management / safety-stock policy; data-integrity program behind the plan

Honesty label. I owned the definitions, the data-integrity program and the delivery; the client's SAP teams and our integration engineers built the feeds; planners owned the forecast — I owned its measurement and the release rule. I have not owned a demand forecast end-to-end and won't claim it. The safety-stock formula is how I explain it now with the MIT toolkit — an illustration, not a program artifact.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. A six-site manufacturer — mixed make-to-order and make-to-stock — learned about a component shortage when the line stopped. S&OP ran off private spreadsheets, shipment data was 60% complete, and supplier, material, BOM, inventory, production, transportation, sales-order and planning data lived in SAP and site systems but never met. Leadership's question — which customer orders are at risk this quarter? — had no data path, so the plan responded with expedites instead of re-plans and buffers sized on noise. **Task.** Nobody owned the seam; I took the definitions, the data-integrity program and the delivery. **Action.** I defined supply honestly before any screen — on-hand plus production plan plus released planned orders only, at SKU × location × date — and refused to count unfirmed MRP orders because they change every run. Domains went live in waves sequenced by planner trust, each behind a published data-quality gate, with daily reconciliation to the SAP source and false alarms worked as defects. The shortage alert fires when days-of-cover falls to the lane's measured p90 lead time — the last moment a re-plan is cheaper than a flight. Safety stock came out only under a rule Planning and Finance signed: release where forecast error and lead-time error had both improved. **Result.** Primary metric: forecast accuracy on the scoped class-A SKUs 70% → 85%. $2M of safety stock released under the two-signal rule — record, one-time working capital — with ≈$0.8M a year of modeled recurring value around it; the shortage alert live in three months; planner effort −50%. That $2M is a different program from the lane program's $1.5M; I never add them.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — A strategic manufacturing customer with six sites: planners learned about a component shortage when the line stopped; S&OP ran off private spreadsheets; shipment data was 60% complete; eight data domains lived in SAP and site systems but never met. Leadership's question — "which customer orders are at risk this quarter?" — had no data path.

The Plan process underneath was running on an unsafe definition of supply (unfirmed MRP orders counted because they looked better), an unmeasured lead-time error, and a shortage signal that arrived at line-down. The cost: expedites instead of re-plans, buffers sized on noise, and planners spending ~40% of their week reconciling rather than root-causing.

T — Nobody owned the seam — no spec, no agreed grain, no definition of "supply," no data-quality bar. I chose all four, as owner of the definitions, the data-integrity program and the delivery. Goal: a real-time supply/demand view with honest definitions and a shortage visible a lead-time early — and planner trust, which was the real product.

Baselines on the table: forecast accuracy 70% on scoped class-A SKUs · shortage detected at line-down (target: a lead-time early) · shipment-data completeness 60% · planner effort on reconciliation and firefighting (≈40% of the week, illustrative share; the −50% is record).

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (4–6 wks)** — profiled every domain (completeness, latency, joinability) to my own gap templates; wrote the definitions memo — grain SKU × location × date; supply = on-hand + production plan + released planned orders only — and the metrics tree (OTIF, inventory $, MAPE/bias, lane lead-time accuracy, completeness, latency, adoption). Refused Planning's request to count unfirmed MRP orders as supply. *Exit bar:* signed gap findings per domain; scope frozen in writing; planners agree the supply definition before any screen.
- **Plan & Design (parallel)** — master-data foundation plan; MPN mapping worklist with its coverage gate; domain-wave sequence by trust; DQ scorecard spec; the 10-step onboarding playbook. *Exit bar:* master-data gates green; unmapped-MPN list under threshold with a named owner; wave order signed by domain owners — who could fail their own wave.
- **Build & Test (8–12 wks)** — domain waves behind published DQ gates — inventory and production first (numbers planners already believed), then purchasing/ASN, then transportation; shortage-alert rule specified (days-of-cover vs lane lead-time p90); daily reconciliation to MD04/COOIS/VA05/MB52; weekly root-cause loop with false alarms worked as defects. *Exit bar:* per-domain completeness ≥ gate; reconciliation passing daily; false alarms trending down; pilot planners using the alert.
- **Cutover & Go-Live (4 wks)** — views live for pilot planners with the at-risk list replacing the Monday spreadsheet; nobody mandated off the spreadsheet — rollback stayed free. *Exit bar:* planners answer "where is my product" from the platform in the live S&OP meeting.
- **Hypercare / Close (4 wks)** — exception workbench staffed; weekly root-cause reviews handed to domain owners; the safety-stock release rule signed with Planning and Finance — release only where forecast error AND lead-time error both improved. *Exit bar:* blockers/day under threshold two consecutive weeks; the release survives audit.

Influence and ownership. Mine: the definitions memo, the metrics tree, the DQ scorecard spec, the wave sequence, the release rule and the onboarding playbook. The client's SAP teams and our integration engineers built the feeds; planners owned the forecast — I owned its measurement (MAPE/bias per SKU family) and the release rule; domain owners held sign-off over their own gates. The MPN worklist gated longer than planned — escalated to the sponsor as a re-sequenced date with the coverage number attached, not a slip.

> **Designed the process; others own it now.** I designed the definitions, the DQ gates, the release rule and the onboarding playbook; domain owners now run the gates and the weekly root-cause reviews, and planners run the alert without me. **Trade-off put to leadership.** The recommendation to Planning and Finance outlined the competing metrics — working capital released vs service risk — and resolved it as a rule: release only where forecast error and lead-time error both improved.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — $2M safety stock released (record, one-time working capital, under the both-errors-improved rule signed with Planning and Finance) + ≈$0.8M/yr *modeled* recurring (carry ≈$0.2M at 10% · expedites pre-empted ≈$0.3M · planner capacity ≈$0.3M; range $0.5–1.2M) — never added to S8's $1.5M.
- **#** — eight data domains into one view across six sites; shortage-alert product live in 3 months; adoption 8× (record).
- **%** — forecast accuracy 70% → 85% on scoped class-A SKUs (record); planner effort −50% (record); shipment data 60% complete at the start.
- **Time** — the alert fires at days-of-cover ≤ lane lead-time p90 — a lead-time early instead of at line-down (record mechanism); views live in 3 months (the alert product; the six-site ingest is a separate 5-month program — never paste 3 months onto it).

- **Customer** — "which customer orders are at risk this quarter?" answered from the platform in the live S&OP meeting, not a spreadsheet (record exit bar).
- **Quality** — daily reconciliation to MD04/COOIS/VA05/MB52; false alarms classified weekly to root cause and burned down as defects (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Eight data domains, six sites and no shared definition of supply. I wrote the definitions memo (grain SKU × location × date; supply = on-hand + production plan + released planned orders only) and the metrics tree, and chose the domain owners who would sign their own gates.
Scope & influence	The client's SAP teams and our integration engineers built the feeds; planners owned the forecast; Planning and Finance signed the release rule; domain owners held sign-off over their gates — none reporting to me.
Execution trade-off + mechanism others now own	Working-capital release vs service risk: release only where forecast error AND lead-time error both improved. Mechanisms others own now: the DQ scorecard, the 10-step onboarding playbook and the weekly root-cause reviews handed to domain owners.
Impact (long-term)	$2M of safety stock released (record); forecast accuracy 70% → 85% on class-A; planner effort −50%; adoption 8×; the shortage alert fires a lead-time early instead of at line-down.
Process improvement (standard work institutionalized)	Domain waves behind published DQ gates; daily reconciliation to MD04/COOIS/VA05/MB52; false alarms classified weekly to root cause and burned down as defects.
Complexity & consensus	Nobody was mandated off the Monday spreadsheet — rollback stayed free until planners answered "where is my product" from the platform in the live S&OP meeting; the MPN worklist gated longer than planned and the gate held.
Root cause at scale	Unfirmed orders inflating supply was a definition defect, not a planner habit — fixed once in the supply definition and applied across all six sites.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **Analytics — How was 70 → 85% measured, and on what?**
>
> Accuracy = 1 − MAPE on the scoped class-A SKUs, with bias published alongside it from month two, at the SKU-family grain the planners decide on. The planners owned the forecast; I owned the measurement and the input integrity — I have not owned a demand forecast end-to-end and won't claim it.

> **Analytics — Why exclude unfirmed MRP planned orders from supply? That understates it.**
>
> Deliberately. Unfirmed orders breathe every MRP run; counting them amplifies noise and inflates supply, which is exactly how the line-down surprises happened. Released planned orders only — accepted by planners before the first screen was built.

> **Analytics — The $2M — how do I know it is not a prettier average?**
>
> Because of the release rule: buffer came out only on SKU families where forecast error and lead-time error had both demonstrably improved, signed by Planning and Finance. That is why the release survived audit. It is one-time working capital; the recurring value around it is my model and labeled.

> **Analytics — Your expedite row is the softest. Defend it.**
>
> Agreed — about sixty expedites a year pre-empted at a $5K premium, both illustrative. The record is the mechanism: the alert fires at days-of-cover ≤ lane p90. To firm it I would pull the expedited-freight ledger for the six sites twelve months before and after the alert went live and count the premium directly.

> **Analytics — Why p90 for the alert threshold and p99 for the promise (S8)?**
>
> Different decisions. The alert is a re-plan trigger — p90 gives the planner a window before the tail bites. The promise is a commitment — it sits at the tail you can defend. Same measured distribution, two policies read off it.

> **Analytics — Define your two error metrics: MAPE versus bias. Which one steered?**
>
> MAPE is mean absolute percentage error — how wrong the forecast is on average, direction ignored; bias is the signed mean error — whether it runs systematically high or low. MAPE without bias hides a chronic over-forecast inside a decent average, and a chronic over-forecast is exactly how buffer quietly accumulates. So accuracy was reported as 1 − MAPE on the scoped class-A SKUs at the SKU-family grain, with bias published alongside from month two — and the release rule required forecast error and lead-time error both to improve, so neither metric could carry the decision alone.

> **Shadow / closing — What would you do differently?**
>
> Start the MPN-mapping worklist on day one — it is never "done" and it gated more than I planned. And define supply honestly before any screen, which held.

THE KEY MATH — THE NETTING WALK AT FINISHED-GOODS GRAIN, THE ALERT RULE AND THE TWO-SIGNAL RELEASE (ILLUSTRATIVE VALUES; RECORD PERCENTAGES)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Accuracy before / after	`1 − 0.30 = 70% → 1 − 0.15 = 85%`	record	70% → 85%
2	Honest supply	`supply(t) = on-hand + in-transit ≤ t + production ≤ t + RELEASED planned orders`	record rule	unfirmed MRP orders excluded
3	Projected balance (FG SKU)	`PAB(3) = 400 − 500 = −100; PAB(7) = 1,000 − 1,200 = −200; PAB(10) = 1,800 − 1,600 = +200`	illustrative FG example — timing shortage, not volume	fix window = day 9 − day 7 = 2 days
4	Alert rule	`DoC = 1,200 / 100 = 12 days ≤ lane p90 14 → alert; p90 = 9 → no alert`	record design; values illustrative	a lead-time early, no false alarm on the fast lane
5	Two-signal release	`SS = z·d·σ_L = 2.33 × 100 × 4 = 932 → σ_L 2: 466; release 466 units — only if BOTH errors improved`	toolkit illustration; the rule is record	release survives audit
6	Recurring value	`$2M × 10% = $0.2M · 60 × $5K = $0.3M · 2.4 FTE × $120K = $0.3M`	modeled (benchmarks labeled)	≈$0.8M/yr; $2M one-time record

MRP math at finished-goods grain only — no BOM explosion in this packet. The six-site ingest (S7) is a separate program; never paste this story's "3 months" onto it.

Where this lands at Amazon. Placement rests on two error signals — forecast error and lead-time error — measured at the decision grain. The transferable disciplines: define supply before the screen, alert at the last moment a re-plan is cheaper than an expedite, and release buffer only when both error signals improve. Reliability (shortage seen a lead-time early), cost efficiency (expedites pre-empted), asset utilization ($2M of working capital released on evidence).

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S6

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you had to dig into the data to find the real problem.	Dive Deep	The supply definition — unfirmed orders inflating supply — found by profiling eight domains; MAPE and bias at the grain planners decide on.	Skip the SAP table names unless asked; systems are enablers.

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you released inventory / improved working capital.	Frugality / Deliver Results	$2M under the two-signal rule — record, one-time; the recurring ≈$0.8M labeled modeled.	Never add the $2M to S8's $1.5M; never claim $2M per year.
Tell me about a time you were right when the business disagreed.	Are Right, A Lot	Refusing unfirmed MRP orders as supply because it looked better; refusing a blanket safety-stock release — agreed as a rule instead.	Do not make Planning the villain; the rule did the refusing.
How do you get planners to trust a new signal?	Earn Trust	Waves sequenced by trust (inventory and production first); false alarms worked as defects; nobody mandated off the spreadsheet; domain owners could fail their own wave.	Do not claim adoption dollars; adoption 8× is the record.
How would you design a shortage / at-risk-order signal for the network?	Functional — Plan process design	Grain, honest supply, days-of-cover vs measured lane p90, DQ gate per feed, two-signal release — the §3.99 opening: book of record and metric first.	No BOM explosion; finished-goods grain only.
What's the softest number in your model?	Highest Standards / Earn Trust	The expedite row (60 × $5K); how I would firm it from the expedited-freight ledger.	Never present planner capacity as cash — it is time redeployed.

S8 — Elementum — replacing brochure lead times with measured lane distributions: promise at p99 on 200 lanes, 99% lead-time accuracy, 580K units / ~$1.5M of safety stock released with zero stockouts

LPs this story can serve here: Dive Deep (primary — measured the distribution, not the average) · Are Right, A Lot (backup — measured vs brochure; p99 on the asymmetry) · Frugality (the release) · Insist on the Highest Standards (the 99% bar before any cut) · Earn Trust (Finance signed each lane; the exception written into the rule)

Process family: Plan / Source / Deliver seam — inbound lead-time policy, safety-stock sizing, promise policy; parameter governance with a drift flag

Honesty label. I owned the requirement (measured lanes, not brochures), the p99 promise policy and the release rule; the data team built the distributions; planners sized the buffers; Finance signed the release. Lane-cohort counts and week markers are not in the record. The safety-stock formula is the MIT inventory-theory toolkit as I explain it now — an illustration, not a program artifact. Depends on the six-site canonical model (S7 — pointer only).

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. The same six-site manufacturer planned inventory on carrier-brochure lead times — every buffer sized on a variability nobody had measured: too much on reliable lanes, too little on fat-tailed ones. The logistics manager asked for a "lead-time variability report." **Task.** I reframed it as a parameter and policy problem — nobody owned the seam between transportation reality and planning parameters, and I took it — scoped at 200 lanes by site, lane and mode. **Action.** I wrote the requirement to the data team: mean, spread and tail per lane from actual production cycle plus inbound transit; defined the promise at p99 cumulatively — the smallest day d with $P(L \le d) \ge 0.99$, not average-plus-padding — and showed each lane's asymmetry, line-down cost against carrying cost, and let planning choose the percentile; they chose p99. The release rule was written as a signed planning act with the exception inside it: fat-tail lanes keep their buffer until the source process moves; Finance agreed the rule in the same room, before any cut. I would not endorse a release until lead-time accuracy hit 99% on the 200-lane set; the stockout-and-expedite watch was armed first; every parameter got a drift flag and a refresh cadence so the promise never decays into a new brochure. **Result.** Primary metric: lead-time accuracy on 200 lanes — brochure values → 99% at the p99 promise. 580K units, about $1.5M of safety stock, released without a stockout — record, one-time working capital — with ≈$0.6M a year modeled around it. A different program from the $2M forecast release; never added.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The six-site manufacturer's logistics manager asked for a "lead-time variability report." The real problem: inventory planning ran on carrier-brochure lead times — every buffer sized on a variability nobody had measured; too much buffer on reliable lanes, too little on fat-tailed ones; the promise to the plan was folklore.

The risk cut both ways: keep planning on brochures and you hold too much on reliable lanes and stock out on fat-tailed ones; cut buffer without evidence and the stockout is yours. And an unmaintained promise decays into a new brochure — which is why every parameter needed a drift flag.

T — Nobody owned the seam between transportation reality and planning parameters; I took it. Goal: measured lanes replace catalog values; a promise at a defensible tail; buffer released only on evidence, with service held — scoped at 200 lanes by site, lane and mode.

Baselines on the table: lead-time parameters = carrier-brochure values, variability unmeasured · buffer capital sized on catalog values · no lead-time-accuracy measurement · target 99% lead-time accuracy on the 200-lane set before any release.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — reframed the report request as a parameter problem; scoped 200 lanes by site/lane/mode; wrote the success criteria — measured lanes replace catalog values; buffer released only on evidence; fat tails keep buffer; showed one right-skewed lane where the average hid the tail. *Exit bar:* planning and logistics agree the problem is the parameter, not the report.
- **Plan & Design** — requirement to the data team (mean, spread, tail per lane; production cycle + inbound transit vs actual); the p99 promise policy (cumulative, not μ + 2.33σ); the release rule as a signed planning act with the fat-tail exception inside it; the stockout-and-expedite watch; drift flag and refresh cadence. On p95 vs p99, showed the asymmetry per lane and left the choice to planning. *Exit bar:* policy signed by planning; Finance agrees the release rule in the room; data team accepts the spec.
- **Build & Test** — data team built the lane distributions and tracked lead-time accuracy per lane; manual reconciliation stayed alive. *Exit bar:* lead-time accuracy 99% on the 200-lane set — the bar I would not endorse a cut below.
- **Cutover & Go-Live** — planners executed the buffer release lane by lane under the signed rule; fat-tail lanes excluded until their source process moved; watch armed first. *Exit bar:* 580K units / ~$1.5M released; no stockout in the watch window.
- **Hypercare / Close** — read the watch; set drift flags per lane; handed the refresh cadence and the release rule to planning as an act they could re-apply. *Exit bar:* watch clean; the promise stays measured.

Influence and ownership. I owned the requirement, the promise policy and the release rule; the data team built the distributions and tracked accuracy; planners sized the buffers and executed; Finance signed each lane. When Finance pushed for a broader release, the rule they had signed did the refusing — fat-tail lanes keep buffer until the source process moves. The percentile fight never happened: I showed each lane's line-down cost against carrying cost and planning chose p99 — their choice surviving an audit is worth more than my choice winning a meeting.

> **Designed the process; others own it now.** I designed the promise policy and the release rule as a signed planning act; planning now re-applies it on its own refresh cadence with a drift flag per lane. **Trade-off put to leadership.** The recommendation to Finance and Planning outlined the competing metrics — ~$1.5M of working capital out vs stockout exposure at the tail — and set the bar: p99, 99% lead-time accuracy, fat-tail lanes excluded until their source process moved.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — 580K units / ~$1.5M of safety stock released without a stockout (record, one-time working capital); ≈$0.6M/yr *modeled* recurring around it (carry ≈$0.3M at a benchmark 20% all-in rate + expedites avoided ≈$0.3M; range $0.3–1.4M) — a different program from S6's $2M; never added.
- **#** — 200 lanes measured by site, lane and mode; 580K units released (record); ≈$2.59/unit derived — low-value components.
- **%** — lead-time accuracy 99% on the 200-lane set at the p99 promise — the bar below which I would not endorse a cut (record).
- **Time** — release only after the 99% bar; watch window clean; refresh cadence and drift flag per lane handed to planning (record).
- **Customer** — service held — zero stockouts on released lanes through the watch window (record).
- **Quality** — release only under the signed rule; fat-tail lanes kept buffer until the source process moved; the promise stays measured, never a new brochure (record).

L6 LEVEL EVIDENCE — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	A report request with no agreed parameter behind it. I reframed it as a parameter problem and wrote the success criteria myself: measured lanes replace catalog values; buffer released only on evidence; fat tails keep buffer.
Scope & influence	200 lanes by site, lane and mode; the data team built the distributions; planners sized and executed; Finance signed each lane — a planning act I designed and others executed.
Execution trade-off + mechanism others now own	Working capital out vs stockout risk: promise at p99 (cumulative), a 99% lead-time-accuracy bar below which I would not endorse a cut, fat-tail lanes excluded until their source process moved. Mechanism others own now: the release rule as a signed act, the refresh cadence and a drift flag per lane handed to planning.
Impact (long-term)	580K units / ~$1.5M of safety stock released without a stockout (record, one-time working capital); the promise stays measured, never a new brochure.
Process improvement (standard work institutionalized)	Lead-time accuracy tracked per lane; drift flags; the release rule written so planning can re-apply it without me.
Complexity & consensus	Finance pushed for a broader release — the rule they had signed did the refusing; the fat-tail exception was inside the rule, so it fired as policy, not as a judgment call.
Root cause at scale	Brochure lead times were the parameter error behind every excess buffer — fixing the parameter fixed all 200 lanes at once rather than one planner's buffer at a time.

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

Analytics — Why p99 and not p95? That is a lot of buffer.

Per lane, the cost of a line-down on a fat-tailed lane dwarfed the carrying cost of the extra day. I showed the asymmetry lane by lane and let planning choose — they chose p99. On reliable lanes the measured σ was so far below the brochure that p99 still released buffer; on fat-tailed lanes the buffer stayed.

Analytics — How did you compute p99 on skewed lanes?

Cumulatively from the empirical distribution — the smallest d with $P(L \le d) \ge 0.99$ — not $\mu + 2.33\sigma$. On one right-skewed lane the normal shortcut understated the true p99 by about 3.4 days; that lane is why the definition is cumulative.

Analytics — Is the $1.5M measured?

The release is record: 580K units out, zero stockouts in the watch window, Finance signed each lane. The ≈$2.59/unit is the derived average. What is modeled is the recurring value around it — carry at 20% all-in (capital-only would be ≈$150K at 10%) and avoided expedites — each input labeled.

Analytics — The expedite number assumes what?

Ten thousand shipments a year on the 200 lanes and one late arrival in ten forcing an expedite before p99 — illustrative. I would pull the expedite ledger by lane for the year before and after and compare directly.

Analytics — What is lead-time accuracy, exactly?

$1 - |\text{actual} - \text{promised}| / \text{promised}$, tracked per lane against the p99 promise; the bar was 99% on the whole 200-lane set. Below it, no release — written into the act before anyone asked.

Analytics — At what grain did you measure lead time, and over what window?

At the lane grain, from actual ship-to-receipt durations — one empirical distribution per lane, never a network average. The record carries the per-lane distributions, the refresh cadence and the drift flag (p99 recomputed on refresh; the buffer decision re-opens if accuracy falls); it does not carry a quoted window length, and I will not invent one — what mattered design-wise was a window long enough to hold seasonality and recent enough that a dead lane's history could not vote, with the drift flag as the safety net either way.

> **Analytics — What does "99% lead-time accuracy" tolerate, exactly?**
>
> The metric is a ratio — 1 − |actual − promised| / promised — so the tolerance scales with the lane instead of being a flat day-count: on a 5-day lane the debit starts at roughly a quarter-day of miss; on a 30-day lane, about a day and a half. A flat day tolerance would have been strict on short lanes and blind on long ones. The bar was 99% across the whole 200-lane set before any buffer released — per-lane measurement, no pooling to hide a bad lane inside a good average.

> **Shadow / closing — How does the promise stay honest after you leave?**
>
> Drift flag per lane with a refresh cadence: p99 is recomputed on refresh, and the buffer decision re-opens if accuracy falls. The stockout-and-expedite watch became a standing report owned by planning.

THE KEY MATH — ONE RELIABLE LANE, ONE FAT-TAIL LANE, AND THE RECURRING VALUE (ILLUSTRATIVE UNITS; RECORD PERCENTAGES AND THE RECORD RELEASE PAIR)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Brochure-sized buffer	`SS = z·d·σ_L = 2.33 × 500 × 5 = 5,825 units`	illustrative; toolkit form	5,825 units of folklore
2	p99 defined cumulatively	`p99 = min{d : P(L ≤ d) ≥ 0.99}`	record definition	normal shortcut understates a skewed lane by 3.4 days
3	Reliable lane re-sized	`2.33 × 500 × 1.2 = 1,398; release = 5,825 − 1,398 = 4,427`	measured σ = 1.2 vs brochure 5 (illustrative)	4,427 units off this lane
4	Fat-tail lane	`2.33 × 500 × 6.5 = 7,573 — buffer KEPT`	measured σ = 6.5 > brochure 5 (illustrative)	kept until the source process moves
5	Accuracy gate	`LTacc = 1 − |actual − promised| / promised; bar 99% on 200 lanes`	record	99% held before any cut
6	The release	`$1,500,000 / 580,000 = $2.59/unit`	record pair; average derived	580K units / ~$1.5M, 0 stockouts
7	Recurring value	`$1.5M × 20% = $0.3M; 900 fewer late × 10% × $3K ≈ $0.27M`	modeled (benchmarks labeled)	≈$0.6M/yr; never added to S6's $2M

Depends on the six-site canonical model (S7) — six definitions cannot produce one σ. S7 is a pointer only in this packet; say "six sites," never "six NA sites."

Where this lands at Amazon. Inbound lead-time variability is half of every buffer and every promise in a placement model. Transferable: measure the lane's own distribution, promise at the tail you can defend, release capital only on evidence with the exception written into the rule, and give every parameter a drift flag. Reliability (zero stockouts), cost efficiency (expedites avoided), asset utilization (~$1.5M of working capital out).

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S8

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you went deeper than the request.	Dive Deep	A "variability report" reframed as a promise-and-buffer policy; mean, spread and tail per lane from actuals; p99 defined cumulatively.	Do not claim to have built the distributions — the data team did, to my requirement.
Tell me about a time the data contradicted conventional wisdom.	Are Right, A Lot	Brochure values vs measured lanes; one right-skewed lane where the average hid the tail; planning chose p99 on the asymmetry.	S10 (Austria — "the noise said carriers") is the bar raiser's; do not use it here.

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you held a standard under pressure.	Insist on the Highest Standards	No release below 99% lead-time accuracy; Finance's push for a broader release answered with the rule they signed.	Do not frame Finance as the obstacle; the exception was in their rule.
How do you set safety stock / service levels?	Functional — placement policy	Two error signals (forecast, lead time), measured at the lane × SKU grain; policy by cost asymmetry; release by rule with a watch window; drift flags.	Never present the formula as a program artifact — MIT toolkit illustration.
What did you refuse to do?	Have Backbone	Averaging away a tail; endorsing a cut below 99%; a blanket release — all written into the act before anyone asked.	Keep the $2M (S6) out of this answer.
How did you make Finance comfortable releasing capital?	Earn Trust	Finance agreed the release rule in the same room as planning signed the policy — before any cut; signed each lane; the watch armed first.	No annualized $1.5M — it is a one-time release.

Question bank — likely questions for this interviewer, with a one-line answer plan

Behavioral and functional questions a Placement Analytics manager is likely to ask. Each plan points at a packet story or a named backup in the master packet; stories allocated to other interviewers are pointers only.

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
1	**Walk me through how you improved forecast accuracy.**	S6 pitch — honest supply definition first, MAPE/bias at the decision grain, waves by trust; 70 → 85% on scoped class-A SKUs; planners owned the forecast, I owned its measurement.	S6
2	**Tell me about a time you used data to change an inventory policy.**	S8 pitch — brochure → measured lanes, p99 promise, 99% bar, 580K units / ~$1.5M out with zero stockouts.	S8
3	**How do you decide where / how much safety stock to hold?**	Two error signals measured at the grain; cost asymmetry sets the percentile; release only by signed rule with a watch; fat tails keep buffer.	S8 → S6
4	**What is the softest assumption in your biggest result?**	S6: the expedite count (60 × $5K). S8: the late-arrival share. Say how each would be firmed from a ledger.	S6 / S8
5	**How do you measure lead-time performance?**	Lead-time accuracy = 1 − |actual − promised| / promised against a per-lane p99 promise; drift flag and refresh cadence.	S8
6	**Tell me about a time a stakeholder wanted a number that looked better.**	Planning's request to count unfirmed MRP orders as supply — refused; released planned orders only.	S6
7	**How did you handle a dependency that slipped (MPN mapping)?**	Escalated to the sponsor as a re-sequenced date with the coverage number attached; made MPN a standing metric rather than a task.	S6
8	**How would you decompose a network placement problem (e.g., inbound placement cost vs. speed)?**	Master §3.99 opening: book of record and metric → stocks and flows → broken invariant → design → sequence; then lead-time and forecast error at the lane × node grain as the two levers.	Master §3.99
9	**How do you validate a model before the business acts on it?**	Parallel run against the source (daily reconciliation to SAP; manual sample per cohort); a published gate; the old process stays alive until the match holds.	S6 (S10 pointer — Josyula's)
10	**What is your quantitative background?**	MIT Supply Chain MicroMasters + Supply Chain Network Design certificate — acquired to design what I had been implementing; the formulas are that toolkit, labeled. Never "I build models for a living."	Master §4.1 toolkit line
11	**Tell me about a time you improved a process end to end (Plan → Deliver).**	S6 as the Plan redesign (definition → gate → alert → release rule) with S8 as the Source/Deliver parameter feeding it; the two programs stay separate in dollars.	S6 + S8
12	**Another example of Dive Deep?**	Pointers only: S10 (Austria — bar raiser's) and S7 (phantom math — unallocated; no BOM-explosion content). If pressed, S7 at the level of "six sites, three waves, five months, zero auto-loaded revision overlaps."	S7 pointer

Cheat-numbers box — canon values with basis labels

Canon numbers for S6 and S8 with their basis labels, from the master packet's §4.1 ledger.

METRIC	VALUE TO SAY	BASIS	NEVER
S6 forecast accuracy	70% → 85% on scoped class-A SKUs	record	claiming I owned the forecast
S6 safety stock	$2M released under the both-errors-improved rule	record, one-time	"$2M per year"; adding to S8
S6 recurring	≈$0.8M/yr (carry 0.2 · expedites 0.3 · planner capacity 0.3); range $0.5–1.2M	modeled	presenting as record
S6 timing / scope	shortage alert live in 3 months; eight domains; six sites; adoption 8×; planner effort −50%; shipment data 60% complete at start	record	pasting "3 months" onto the six-site ingest (S7)
S6 alert rule	days-of-cover ≤ lane lead-time p90	record mechanism	—
S8 lanes / accuracy	200 lanes; 99% lead-time accuracy at the p99 promise	record	—
S8 release	580K units / ~$1.5M, zero stockouts in the watch window	record, one-time	"$1.5M to $2M"; annualizing it
S8 recurring	≈$0.6M/yr (carry 0.3 at 20% · expedites 0.3); range $0.3–1.4M	modeled	presenting as record
S8 unit value	≈$2.59/unit	derived from the record pair	—
Site count	"six sites"	record	"six NA sites"
Toolkit	MIT Supply Chain MicroMasters + Network Design certificate; formulas are illustrations	labeled	"I build supply chain models for a living"
S7 (pointer only)	six sites, three waves, 5 months; hardest site 6 weeks vs 3-week average	record	"500+ product families"; 60→95% inside S7; any BOM-explosion walk

Never-say, one glance before the call.

- Never add $2M (S6) and ~$1.5M (S8) into one number; never "$1.5M to $2M."
- Never "six NA sites" — "six sites." Never paste S6's "3 months" onto the six-site ingest.
- Never claim ownership of the demand forecast or the distributions — measurement, definitions, rule and policy were mine.
- Never present the safety-stock formula as a program artifact; MRP math at finished-goods grain only — no BOM explosion.
- Never present planner capacity as cash — time redeployed, not headcount removed.
- S10 (Austria) is the bar raiser's; S11/S12, S2/S18, S1/S3 belong to Tripathy, Goforth and Lan — pointers only here.

Backup bench — use only if a primary doesn't fit

Four fallbacks for an analytics manager: three master-packet backups plus one doc-derived Flexport micro-story — S4 (Sephora — the in-transit promise rule I got wrong; the failure / Are Right, A Lot question, promise math at lane σ), S9 (Elementum — the 597-lane spend Pareto and the ≥95% valid-ASN gate; Dive Deep on data completeness, Backbone two levels up) and S7 (Elementum — a contract manufacturer's planning data made safe to plan on, finished-goods grain only; Learn & Be Curious by math, not meetings); the doc-derived fourth is the FFA Asia latency investigation from the Flexport program record (Dive Deep on production evidence; record-from-docs labels, facts identical to the same bench in Josyula's packet). Tell S6 and S8 first.

Bench discipline. The two (or four) primaries above are the stories to tell first. Reach for a bench story only when the question genuinely does not fit a primary, or when the interviewer asks for "another example" on an LP already spent. Each entry is a 60-second pitch, a compressed STAR, one results line and two probes — enough to answer and stop; no full probing bank or derivation. Where a bench story is another interviewer's primary, the red note says so: use it only if the question demands it, and keep the facts identical to that packet.

BACKUP — ONLY IF ASKED S4 — Sephora S/4HANA — the in-transit promise rule I got wrong, fixed through the signed policy, not a side configuration

LPs this backup can serve: Are Right, A Lot (the failure question) · Customer Obsession (backup) · Earn Trust (owning the miss) · Learn and Be Curious — the designated policy-failure story

Basis. The v1 rule was mine; the indictment came from my own program's reports. Lane σ ≈ 0.3 day and the ~0.49-day window are the record's promise math; the record carries no lane counts and no dollar for the conversion lost while I was wrong — I will not invent the loss figure.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. My first ISM ruleset barred every channel from promising units in transit; the Aging In-Transit report showed dotcom starved of fast movers one day from the building. **Task.** Correct my own policy without weakening oversell protection, through the same three signatories. **Action.** Discovery: conversion-loss evidence from the aging and replenishment reports — the entry bar to reopen the policy; I stated that the v1 rule was mine. Plan & Design: promise math — lane σ ≈ 0.3 day, a ~0.49-day window ≈ 95% coverage; Promisable = in-transit × 1[ETA ≤ window] × 1[on-time ≥ floor]; Aging In-Transit as the kill switch; both blanket options rejected in writing. Build & Test: back-test against receiving history; four regression cells added to the node × channel matrix. Cutover: released on the normal monthly cadence as a signed FSD change, lane by lane — no hotfix, no side configuration. Hypercare: kill switch armed; never fired. Lesson: both sides of a rule get a number before it ships. **Result.** Primary metric: sellable units on qualifying in-transit lanes — starved → restored inside a per-lane promise window, oversell increase = zero, kill switch never fired. ≈$0.5M/yr modeled margin recovered — the record has no dollar for the conversion lost while I was wrong, and I will not invent one.

COMPRESSED STAR

- **S —** v1 of my own ISM ruleset: no channel could promise in-transit units — safe-feeling, and starving dotcom of fast movers reliably one day from the building.
- **T —** Fix my own signed policy without weakening oversell protection; take the miss back through the same three signatories.
- **A —** Evidence pack from Aging In-Transit and Replenishment; reframed as promise math (σ ≈ 0.3 d → ~0.49-d window ≈ 95%); rule with on-time floor and kill switch; back-tested against receiving history; four regression cells; released lane by lane on the monthly cadence as an FSD change.

- **R —** Starved lanes restored inside the window; oversell increase = 0; kill switch never fired; re-signed by the same three signatories; lesson institutionalized — the Locus 98% gate (S2) and the 0.5% tolerance (S18) were set from data before the event.

RESULTS LINE

Sellable units on qualifying in-transit lanes starved → restored inside a back-tested ~0.49-day window (record) · oversell increase = 0; kill switch armed, never fired (record) · four regression cells added (record) · released as a signed FSD change on the normal cadence, no hotfix (record) · ≈$0.5M/yr modeled margin recovered (inputs illustrative).

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	The v1 rule	`Promisable = 0 for ALL in-transit inventory`	equivalent to claiming σ → ∞ or on-time rate → 0 — infinite variability (record)	everything in transit hidden
2	One concrete lane	`Promisable = 120 × 0 = 0`	lane with InTransit = 120 units, ETA 0.4 day away (illustrative)	120 sellable units hidden
3	Across qualifying lanes	`≈300 fast-mover units/day × 0 promisable`	illustrative volume	≈300 units/day sat unsellable one day from the building
4	What starving costs (one SKU)	`100 × 60% × $20 = $1,200/day`	100 units one day out; 60% would convert; $20 margin (illustrative, not program data)	$1,200/day under-sell
5	The ignored other side	`100 × (1 − 0.97) = 3 late units; 3 × $15 = $45/day`	on-time rate 97%; $15 expedite (illustrative)	$45/day over-sell exposure — vs $1,200: the “safe” rule priced one side only

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Measure lane variability	`σ = 0.3 day`	trailing receiving history (record)	σ = 0.3 day
2	Choose coverage	`z at 95% = 1.645`	normal approximation on lane transit (toolkit label)	z = 1.645
3	Promise window	`w = z × σ = 1.645 × 0.3 = 0.4935`	rows 1–2	w ≈ 0.49 day
4	Coverage check	`Φ(w / σ) = Φ(0.49 / 0.3) = Φ(1.63) ≈ 0.95`	share of arrivals inside the window	≈95% covered
5	Corrected rule	`Promisable = InTransit × 1[ETA ≤ w] × 1[OTR ≥ floor]`	OTR = on-time rate over trailing receipts; floor = per-lane admission threshold, 95% here (floor illustrative)	lane by lane, not all-or-nothing
6	Lane that qualifies	`ETA 0.4 ≤ 0.49 → 1; OTR 97% ≥ 95% → 1; Promisable = 120 × 1 × 1 = 120`	the Table 1 lane, re-evaluated (ETA and OTR illustrative)	its 120 units become promisable
7	Lane that fails	`ETA 1.2 > 0.49 → 0; Promisable = 80 × 0 = 0`	a slower lane with 80 units in transit (illustrative)	stays at 0 — correctly out
8	Sample-size gate	`n = z²p(1−p)/e² = 3.84 × 0.95 × 0.05 / 0.05² = 3.84 × 0.0475 / 0.0025 ≈ 73`	OTR estimated within ±5% at 95% CI (illustrative)	a lane qualifies after ~73 receipts

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
9	Kill switch, values in	`trigger = 2 × p90 = 2 × 2.0 = 4.0 days; any unit in transit > 4.0 days → lane out`	lane p90 = 2.0 days (illustrative); armed, reviewed weekly by Inventory Control (record)	never fired (record)
10	Back-test	`rule replayed against receiving history; 4 regression cells`	before release (record)	no oversell in the back-test

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Sellable units on qualifying in-transit lanes	0	≈300 units/day restored inside the window	record mechanism; volume illustrative
2	Promise window	none — blanket exclusion	w ≈ 0.49 day at ~95% cover	record (promise math)
3	Oversell increase from the window	—	0	record
4	Kill switch	not armed	armed, never fired	record
5	Rule changes after	ad hoc	same 3 signatories, back-test + matrix regression	record — both sides priced from then on

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Would-be sales	`300 × 60% = 180 sales/day`	units restored (illustrative) × conversion (illustrative)	180 sales/day
2	Truly lost without the fix	`180 × 40% = 72 sales/day`	40% lost-sale share — the rest would have waited (illustrative)	72 lost sales/day
3	Margin recovered	`72 × $20 = $1,440/day`	$20 unit margin (illustrative)	≈$1,440/day
4	Annualized	`$1,440 × 365 = $525,600`	modeled; the mechanism is record	≈$0.5M/yr
5	Cost while v1 stood	`no dollar in the record — not claimed`	record	not claimed

TWO PROBES

> **Why not just hotfix it — you knew it was wrong?**
>
> Because the error was in the policy, so the fix had to be too — in the governed artifact, with regression, through the same signatures. A side configuration would have fixed the symptom and left the policy lying about what it allowed.

> **What did the miss cost?**
>
> The record carries no dollar for the conversion lost while v1 stood, and I will not invent one. What I can say is the mechanism: fast movers reliably one day from the building were unsellable on dotcom until the window existed; the ≈$0.5M/yr recovered is my model on illustrative inputs, labeled.

Canon numbers (basis labels): Lane σ ≈ 0.3 d; ~0.49-d window ≈ 95% coverage (record promise math) · oversell increase = 0 (record) · 4 regression cells (record) · kill switch never fired (record) · same three signatories re-signed (record) · ≈$0.5M/yr (modeled). Never a lane count; never a dollar for the loss while wrong.

BACKUP — ONLY IF ASKED S9 — Elementum — 1,000 supplier/carrier routes asked, 597 built first: X12 EDI onboarding behind a ≥95% valid-ASN cohort gate

LPs this backup can serve: Have Backbone; Disagree & Commit (primary — 597 vs 1,000, two levels up) · Deliver Results · Insist on the Highest Standards (the gate held twice) · Think Big (the staged 1,000-lane vision) · Broad Responsibility (small carriers kept inside the standard by the portal floor)

Basis. I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor; engineers built the maps and harness to the standard; Transportation owned the carrier relationships. Cohort sizes and the number of cohorts are not in the record — do not quote them.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. The Global Head of Supply Chain wanted inbound visibility on 1,000 lanes in two months; shipment-data completeness was 60% and high-value lanes were blind. **Task.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor. **Action.** Discovery: a spend Pareto — 597 lanes carried 80% of freight spend — pre-validated with Transportation and Procurement before I took it up. Plan & Design: the onboarding standard (X12 850/855/856/940/945/214/997 with an XML/portal floor — "the standard was the data, not the transport") and a cohort gate of ≥95% valid ASN. Build & Test: a mapping-and-test harness and a scorecard per lane. Cutover: 597 live in month one; one cohort failed the gate twice on a timestamp format and I held it both times. Hypercare: completeness scorecards to 95%; ~200 lanes the next wave behind the same gate. Lesson: automating dirty data manufactures confident wrong answers — the gate is the product. **Result.** Primary metric: inbound shipment-data completeness 60% → 95%. Freight spend −10% ≈ $2M in six months (record); 597 lanes = 80% of spend live in month one; adoption 8×.

COMPRESSED STAR

- **S —** 1,000 global lanes asked in two months — scope, schedule and quality all fixed; completeness 60%; lanes differed by orders of magnitude in spend and carriers differed in EDI capability.
- **T —** Own the scope analysis and the onboarding standard; take the re-scope two levels up with the evidence attached.
- **A —** Spend Pareto (597 = 80% of spend) pre-validated so no lane owner could contradict it in the room; standard + ≥95% valid-ASN gate published to carriers before cohort one; harness so later cohorts were mostly configuration; held a failing cohort twice; ~400 lanes carrying ~20% of spend deferred — priced in spend share, not lane count.
- **R —** Completeness 60 → 95%; 100% visibility on high-value lanes; 597 lanes live month one; freight −10% ≈ $2M in six months (record); method written up as the sourcing SOP.

RESULTS LINE

Completeness 60% → 95% (record) · 597 of 1,000 lanes = 80% of freight spend live in month one (record) · freight spend −10% ≈ $2M in six months (record — kept as measured, never annualized) · adoption 8× (record) · ≈$0.4M/yr document-handling labor (modeled add-on).

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Completeness	`480 / 800 = 60%`	expected = 200 shipments × 4 milestones = 800; received 480 (illustrative volumes; 60% record)	60% — 2 of 5 events missing

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	What blindness costs	`off-guide spend unseen · premium modes by habit · chase calls per late shipment`	record problem	no instrument, no lever
3	Manual documents	`597 lanes × ~50 shipments × 3 docs ≈ 90,000 docs/yr keyed by hand`	illustrative volumes	~90K docs/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Spend Pareto	`S(k) = cumulative spend share; S(597) = 80%`	lanes sorted by annual freight spend (record)	597 of 1,000 lanes carry 80% of spend — do them first
2	Cohort gate	`validASN rate ≥ 95% in test before activation`	published to carriers before cohort one (record)	one cohort failed twice on a timestamp format — held both times
3	Long-tail floor	`XML / portal when EDI is absent`	“the standard was the data, not the transport” (record)	small carriers kept inside the standard
4	Completeness after	`760 / 800 = 95%`	illustrative volumes; 95% record	95%; high-value lanes 100%

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Shipment-data completeness	60%	95%; high-value lanes 100%	record
2	Lanes live month one	0	597 = 80% of spend	record
3	Freight spend on onboarded lanes	base	−10% ≈ $2M in 6 months	record
4	Document handling	~90K docs/yr keyed	X12/XML validated	modeled

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Implied addressable base	`$2,000,000 / 10% = $20,000,000/yr`	derived from the record pair, stated as such	≈$20M/yr
2	The record result	`$20M × 10% = $2M in 6 months`	record	≈$2M
3	Slice — routing-guide compliance	`~$0.8M of the $2M`	off-guide spend caught by 856/214 (modeled split)	~$0.8M
4	Slice — mode discipline	`~$0.6M of the $2M`	premium modes → ocean/ground when ETA is trusted (modeled split)	~$0.6M
5	Slice — expedites + detention	`~$0.6M of the $2M`	events replace chase calls (modeled split)	~$0.6M
6	Split check	`0.8 + 0.6 + 0.6 = $2.0M`	the slices decompose the record — they never add to it	= the record $2M
7	Document labor	`90,000 × 6 min = 540,000 min; / 60 = 9,000 h; × $45 = $405,000`	illustrative volumes (modeled add-on)	≈$0.4M/yr

TWO PROBES

> **You defied a Global Head of Supply Chain's number. How?**
>
> With the spend curve and the lane owners' validation attached — 597 lanes carried 80% of the spend, so the phased plan was argued in spend share, not lane count. He accepted the phasing off the curve; I then committed to the remainder behind the same gate and delivered it.

> **Is the $2M annualized?**
>
> No — it is the record as measured: minus ten percent on the onboarded lanes' spend in six months. My three-way split (routing-guide compliance ~$0.8M, mode discipline ~$0.6M, expedites and detention ~$0.6M) decomposes the record; it never adds to it.

Canon numbers (basis labels): 60% → 95% completeness (record) · 597 / 1,000 lanes = 80% of spend (record) · −10% ≈ $2M in 6 months (record) · ≥95% valid-ASN gate, held twice (record) · 8× adoption (record) · ≈$0.4M/yr labor (modeled). Never: cohort counts or sizes; never double the $2M into a run-rate.

BACKUP — ONLY IF ASKED S7 — Elementum — making a contract manufacturer's planning data safe to plan on: six sites, three waves, five months, zero auto-loaded revision overlaps

LPs this backup can serve: Learn and Be Curious (primary — learned the CM's data by math, not meetings) · Dive Deep · Earn Trust (external partner)

Basis. I owned the ingest requirements, the reconciliation gate, the wave plan and the weekly Flextronics call; engineers built the normalization layer and quarantine to the spec; the commercial relationship belonged to the customer's procurement — my leverage was evidence. The $1.5M / 580K belong to the p99 program this backbone enabled (S8). Finished-goods grain only here; no tolerance value — the record does not carry one.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. Flextronics supplied the planning data our component-demand visibility depended on, and it carried two traps documentation never confesses: pass-through assemblies that double-time demand if loaded as stocked, and overlapping revisions that rewrite history if you auto-load one. Six site systems carried six definitions of yield, lead time and order status. **Task.** I owned the ingest requirements, the reconciliation gate, the wave plan and the weekly Flextronics call — with no commercial leverage; evidence was my only lever. **Action.** Discovery: sampled finished goods and reconciled the CM's planning data against its own MRP — the trap signature found by math, named in a discovery memo. Plan & Design: an explicit marker for pass-through items; an effective-date quarantine with human disposition — never auto-load; one supply equation (on-hand + production plan + released planned orders) signed by customer planning. Build & Test: mapping tables owned by named site super-users; a per-site reconciliation report as the go-live gate. Cutover: three waves — two cleanest high-volume sites, then three on parameterized maps, then the hostile legacy ERP by snapshot-diffing at six weeks against a three-week average, priced up front. Hypercare: the weekly call left standing — it had become theirs. Lesson: reconcile to the source's math, not its documentation. **Result.** Primary metric: contract-manufacturer sites with planning-grade data — 0 → 6 in three waves and five months, zero auto-loaded revision overlaps. ≈$0.3M/yr modeled — the six sites and zero overlaps are measured; the dollars are my model; the $1.5M belongs to S8.

COMPRESSED STAR

- **S —** A contract manufacturer's planning feed with two silent traps; six sites, six definitions; one hostile legacy ERP (flat files, no change-data-capture); no commercial leverage.
- **T —** Own ingest requirements, the reconciliation gate, the 2/3/1 wave plan, and chair the weekly Flextronics call.
- **A —** Found the traps by sampling finished goods against the CM's own MRP; designed the two mechanisms that made the traps impossible rather than policed (explicit marker; effective-date quarantine, human disposition); reconciliation report as the per-site gate, not a date on a slide; wave 3's six weeks priced to leadership as the plan, not a slip; made the weekly call the CM's own protection against blame for misses its feed didn't cause.
- **R —** Planning-grade sites 0 → 6 in 3 waves / 5 months; zero auto-loaded overlaps ever; phantom double-timing eliminated at source; backbone handed to the 200-lane program (S8).

RESULTS LINE

CM sites with planning-grade data 0 → 6 in three waves / five months (record) · auto-loaded revision overlaps = 0, ever (record) · hardest site 6 weeks vs the 3-week average, priced up front and absorbed in plan (record) · ≈$0.3M/yr (modeled — reconciliation labor, expedites fired at false shortages, overlap rework); the record has no dollar for this story; the $1.5M belongs to S8.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Planning-grade CM sites	`0 of 6; manual BOM-vs-MRP reconciliation at every site`	record	0 of 6
2	Manual recon burn	`6 h/wk × 6 sites = 36 h/wk`	modeled	36 h/wk of reconciliation
3	What MRP computes (finished-goods grain)	`NetReq(FG, t) = GrossReq − OnHand − SchedReceipts`	the netting identity; walk below at FG grain only (illustrative)	substituted in rows 4–5
4	Clean row	`NetReq = 500 − 150 − 200 = 150`	Gross 500; on-hand 150; receipts 200 (illustrative)	NetReq = 150; need-date = due − LT_FG
5	Distorted row — phantom FG record	`NetReq = 600 − 150 − 200 = 250; false shortage = 250 − 150 = 100`	a phantom/stale finished-goods record inflates gross requirements by 100 (illustrative)	an expedite fires on 100 units never needed
6	How often	`~5/site/mo × 6 sites × 12 mo = 360 false shortages/yr`	modeled count	360/yr
7	Timing corruption	`the same stale record pulls the need-date a lead-time early`	record mechanism	planners chase demand that was never real
8	Revision overlaps	`two revisions both partially true in [d0, d1]; ~24/yr`	auto-loading would rewrite demand for orders released under the prior revision (record rule; count modeled)	~24 history-rewrite risks/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Explicit marker	`distorted row reconciled: NetReq = 500 − 150 − 200 = 150`	the phantom record carries an explicit marker; it no longer inflates gross (record mechanism)	NetReq = 150 — matches the site's own MRP
2	Per-site gate	`count_model vs count_MRP on sampled finished goods must match`	the reconciliation report is the go-live gate, not a date on a slide (record)	a site opens on a matched report
3	Quarantine	`overlaps held for human disposition per order/effectivity`	record rule	auto-loaded overlaps = 0 — ever
4	Hostile flat-file site	`completeness = rows received / expected; a 40%-rows file would read as 60% deletions → blocked`	snapshot-diffing with a completeness flag (record mechanism)	partial files never consumed
5	Wave plan	`W1: 2 × 3 = 6 wk; W2: 3 × 3 = 9 wk; W3: 6 wk; 6 + 9 + 6 = 21 weeks`	hostile site at 6 wks vs the 3-wk average, priced up front (record)	≈5 months for six sites
6	Recon labor after	`6 → 1.5 h/wk per site — disposition queue only`	modeled	−4.5 h/wk per site

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Planning-grade CM sites	0 of 6	6 of 6, in 3 waves ≈ 5 months	record
2	Manual reconciliation	6 h/wk × 6 sites	1.5 h/wk × 6	modeled
3	Expedites on false shortages	~72/yr acted on	~0 — the marker removes the distortion	modeled
4	Auto-loaded revision overlaps	uncontrolled	0 — ever	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Reconciliation labor	`(6 − 1.5) = 4.5 h/wk; × 6 sites = 27 h/wk; × 48 wk = 1,296 ≈ 1,300 h; × $60 = $77,760`	modeled	≈$0.08M/yr
2	False-shortage expedites	`360 × 20% = 72 acted on; 72 × $3,000 = $216,000`	count from Table 1 row 6; premium benchmark $2–5K (modeled)	≈$0.2M/yr
3	Overlap rework avoided	`24 × 16 h = 384 h; 384 × $60 = $23,040`	the quarantine makes the error 0 by construction (modeled)	≈$0.02M/yr
4	Total	`0.08 + 0.2 + 0.02 = $0.3M`	the record has no dollar for S7; the $1.5M belongs to S8	≈$0.3M/yr

TWO PROBES

> **How did you get an external supplier with no obligation to care?**
>
> By making the weekly call theirs: it protected Flextronics from blame for misses their feed didn't cause. Procurement held the commercial relationship; my lever was the evidence in the discovery memo, confirmed by their own engineering.

> **Why three waves instead of one?**
>
> So one surprise could not spend six sites of credibility — the two cleanest high-volume sites first, three on parameterized maps second, the hostile ERP last with its six weeks priced up front. Speed was traded, never quality: a site went live off its matched reconciliation report, not a date.

Canon numbers (basis labels): 0 → 6 sites, 3 waves, 5 months (record) · zero auto-loaded overlaps (record) · 6 weeks vs 3-week average (record) · ≈$0.3M/yr (modeled). Never "six NA sites"; never "500+ product families"; 60 → 95% belongs to S9; the $1.5M / 580K belong to S8; no tolerance value.

BACKUP — ONLY IF ASKED Doc bench — Flexport FFA — the Asia latency investigation: a 21-second P95 traced to routing, decomposed to the layer, fixed in sequence (Dive Deep)

LPs this backup can serve: Dive Deep (primary — decomposition to the layer) · Deliver Results · Learn & Be Curious · Insist on the Highest Standards (a smoke-test protocol per change)

Basis. Doc-derived micro-story from the program record (the FFA Asia latency investigation, post-MVP 2022) — every figure is RECORD-FROM-DOCS: the investigation document, written when the FF App MVP had launched to 11 partners in 10 countries and 48 of 53 import partners (90.5%) had migrated. Program-lead voice: I owned the investigation, the decomposition and the fix sequence; infrastructure teams executed the WAF and CDN changes. Not in the master packet's S-numbering; supplements canon, never overrides it. The same bench appears in Josyula's packet — facts identical.

60-SECOND PITCH — STOP ON THE PRIMARY METRIC

Situation. Weeks after the FF App MVP launched to 11 partners in 10 countries — with 48 of 53 import partners already migrated — the monthly business review showed Asia page loads at a P95 of 21.19 seconds against 3.3 in the US. **Task.** Find the layer, not the anecdote, and fix it without a re-architecture in the middle of a live partner-facing launch. **Action.** I decomposed the problem before prescribing: static assets ran ~5s P95 in China against ~500ms in the US, and the app's domain had no geo-based DNS or reverse proxy — Asia traffic was riding the public internet across the Pacific; a double CDN sat on the static path. The fix was a sequence, not a project: enable the WAF, remove the double CDN (direct to S3), then a regional CDN proposal with a smoke-test protocol per change, then China tuning, then Asia tuning — each with a date and an owner. **Result.** Root cause isolated to routing — not the app, not the partners; two fixes shipped inside the investigation window and the regional plan scheduled with dates; and a standing rule imported into every later launch: the pilot's geography is a performance requirement, not a rollout detail. RECORD-FROM-DOCS

COMPRESSED STAR

- **S —** Post-MVP, the MBR surfaced Asia P95 page load at 21.19s vs 3.3s in the US (17-day measurement window); 48 of 53 import partners (90.5%) already migrated — the app had to hold where the partners were.
- **T —** Isolate the layer causing Asia latency and sequence the fix set without destabilizing a live partner-facing launch.
- **A —** Split static from dynamic: static JS ~5s P95 in China vs ~500ms US; found no geo-based DNS / reverse proxy on the domain (public-internet routing) and a double CDN on static assets; sequenced WAF enablement → double-CDN removal (direct to S3) → regional CDN proposal → China tuning → Asia tuning, each gated by a smoke-test protocol with a named date and owner.
- **R —** Root cause isolated to routing; WAF and double-CDN fixes shipped inside the window; CDN / China / Asia work scheduled with owners; the investigation became the template for launch-geography checks.

RESULTS LINE

P95 page load 21.19s Asia vs 3.3s US (record-from-docs) · static ~5s China vs ~500ms US (record-from-docs) · 48 of 53 partners migrated = 90.5% (record-from-docs; canon's "48 of 53 within a month" — consistent) · two fixes shipped, three sequenced with dates (record-from-docs) · no post-fix latency figure exists in the record — never invent one.

THE MATH — STEP-BY-STEP DERIVATION

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Page-load P95 by region	`Asia = 21.19s; US = 3.3s; Others = 4.45s`	Datadog @view.loading_time, 04/24–05/10 window (record-from-docs)	Asia is the outlier

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
2	The gap factor	`21.19 ÷ 3.3 = 6.42`	row 1 (record-from-docs)	6.42× slower than the US
3	Traffic behind the number	`6.31 ÷ 2.19 = 2.88`	Asia 6.31k vs US 2.19k page views, same window (record-from-docs)	≈2.9× the US traffic riding the worst latency
4	Static-asset leg	`JS P95 = 5s China vs 500ms US; 5 ÷ 0.5 = 10`	static.flexport.com CDN behavior (record-from-docs)	10× gap on the static leg
5	Dynamic leg	`no geo-DNS / reverse proxy for trade.flexport.com → Asia GraphQL hairpins over the public internet`	infra review (record-from-docs)	routing penalty on every data call
6	Known defect on the static path	`double CDN in China — Cloudflare stacked in front of Tencent`	infra review (record-from-docs)	one avoidable hop

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Decompose before touching	`page load = static resources + GraphQL requests + JS rendering`	the metric's own definition (record-from-docs)	three legs, measured and owned separately
2	Step 1 — security gate first	`enable WAF for trade.flexport.com: China, then global`	incident-207 follow-up; done 2022/06/30 (record-from-docs)	shipped — fix 1 of 2
3	Step 2 — remove the double CDN	`Tencent CDN → direct to the S3 bucket; Cloudflare hop removed for China`	done 2022/06/24 (record-from-docs)	shipped — fix 2 of 2
4	Step 3 — static proposal, CN/Asia	`alternatives written, reviewed, signed off`	ETA 2022/07/15 (record-from-docs)	sequenced with a date
5	Step 4 — China tuning	`e.g. replicate static assets in-region`	ETA 2022/07/30; depends on step 4 (record-from-docs)	sequenced with a date
6	Step 5 — Asia tuning	`the same playbook beyond China`	ETA 2022/08/15 (record-from-docs)	sequenced with a date
7	Held in reserve	`global CDN for trade.flexport.com — built only if the metrics still demand it`	2022 Q3 or later, metrics-gated (record-from-docs)	no speculative spend
8	Verification per change	`2 user types × 2 instances = 4 smoke passes before + 4 after = 8`	login → workview → logout → re-login; internal SSO + partner users; US + China hosts (record-from-docs; the multiplication counts the checklist)	8 workflow passes around every change

Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Root cause	one number — "Asia is slow"	static leg (10×) and dynamic routing isolated, each with an owner and a sequence	record-from-docs
2	Fixes	none	2 shipped (WAF; double-CDN removal); 3 sequenced with dates (07/15, 07/30, 08/15)	record-from-docs
3	Post-fix P95	21.19s	no post-fix figure exists in the record — never claimed	record-from-docs

STEP	METRIC	BEFORE	AFTER	BASIS
4	Spend discipline	—	global CDN gated on metrics, not built on speculation	record-from-docs

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	INPUTS (BASIS)	VALUE
1	Partner base exposed	`48 ÷ 53 = 0.9057 ≈ 90.5% of import partners already migrated when the gap surfaced`	migration tracker (record-from-docs)	90.5% of partners on the affected app
2	Excess wait served per window	`gap vs US = 21.19 − 3.3 = 17.89s; 6,310 × 17.89 = 112,886 partner-seconds; ÷ 3,600 ≈ 31.4 hours`	6.31k Asia page views in the 17-day window (record-from-docs); gap arithmetic illustrative	≈31.4 partner-hours of excess wait every 17 days
3	Dollar value of the wait	`no dollar in the record — not claimed`	record-from-docs	not claimed
4	Post-fix saving	`no post-fix latency figure exists → no savings chain exists`	record-from-docs	never invent one

TWO PROBES

How did you know it was routing and not the application?

By decomposition before remediation: the static/dynamic split put ~5s of the China time on assets the app server never touched, and the same assets served in ~500ms from the US — the code was constant, the path was not. The domain's missing geo-DNS / reverse proxy and the double CDN explained both halves of the gap without a single application change.

Why not just stand up a China region?

A region is a re-architecture with a compliance tail; the evidence said the bottleneck was routing and CDN topology, which fixes in weeks. The sequenced plan — WAF, double-CDN removal, regional CDN, tuning — bought most of the win at a fraction of the risk, and the smoke-test protocol meant each step was reversible.

Canon numbers (basis labels): 21.19s vs 3.3s P95 (record-from-docs) · ~5s vs ~500ms static (record-from-docs) · 48/53 = 90.5% migrated (record-from-docs) · 11 partners / 10 countries MVP (record — matches S12 canon) · fix sequence WAF → double-CDN → regional CDN → China → Asia with smoke tests (record-from-docs). Never: a post-fix latency number (none in the record); never present scheduled items as delivered.